

CONTENTS

ABSTRACT.....	6
CHAPTER 1 - INTRODUCTION	
1.1 Molecular Spectroscopy.....	7
1.2 Regions of the Spectrums.....	9
1.3 The Vibrations of Polyatomic Molecules.....	10
1.4 Energy Levels.....	11
1.5 Hartree-Fock Theory.....	11
1.6 The Hartree –Fock Approach.....	12
1.7 Basis Set and GTOs.....	13
1.8 Gaussian 09.....	13
CHAPTER 2	
2.1 Literature Review.....	15
CHAPTER 3	
3.1 Computational method.....	16
CHAPTER 4	
4.1 Result and Discussion.....	18
4.2 Molecular Structure.....	18
4.3 IR Spectrum.....	21
4.4 Geometrical Parameters.....	24
4.5 Electric Dipole Moment	25
4.6 Frontier molecular orbitals.....	26
4.7 Electrostatic potential of molecules.....	29
CHAPTER 5	
5.1 Summary and Conclusion.....	30
5.2 References.....	30

ABSTRACT

The spectral and theoretical aspects of selected benzene and aniline molecules are discussed. The IR spectra of these molecules and the vibrational wavenumbers were computed using HF method. The data obtained from theoretical calculations using Gaussian09 were used to assign vibrational bands experimentally. The geometrical parameters and the change when the NH₂ added to the benzene structure were tabulated. The electrostatic potential of the molecule was obtained and visualized. The Highest occupied and Lowest un-occupied molecular orbitals were visualized; HOMO and LUMO energies were calculated. The dipole moment of the molecule also was calculated.

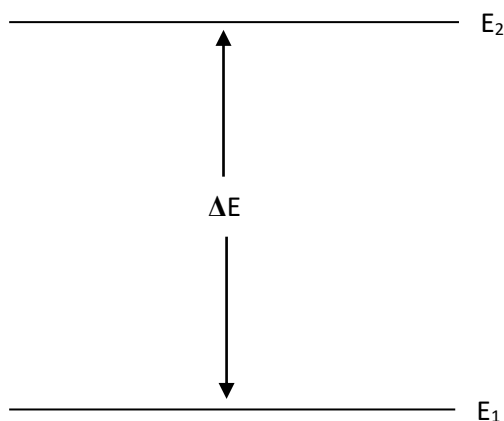
CHAPTER 1

INTRODUCTION

1.1 MOLECULAR SPECTROSCOPY

Molecular spectroscopy may be defined as the study of the interaction of electromagnetic waves and matter. A molecule in space can have many sorts of energy; e.g. it may possess rotational energy by virtue of bodily rotation about its center of gravity; it will have vibrational energy due to the periodic displacement of its atoms from their equilibrium positions; it will have electronic energy since the electrons associated with each atom or bond are in unceasing motion, etc. The chemist or physicist is early familiar with the electronic energy states of an atom or molecule and accepts the idea that an electron can exist in one of the several discrete energy levels: he learns to speak of the energy as being quantized—a particular molecule can exist in a variety of rotational, vibrational, etc., energy levels and can move from one level to another only by a sudden jump involving a finite amount of energy.

Consider two possible energy states of a system—two rotational energy levels of a molecule, for example—labelled E_1 and E_2 in the following diagram:



The suffixes 1 and 2 used to distinguish these levels are, in fact, *quantum numbers*. The actual significance of quantum numbers goes far deeper than their use as a convenient label—in particular, we shall later see that analytical expression for energy levels usually involve an algebraic function of one or more quantum numbers. Transitions can take place between the levels E_1 and E_2 provided the appropriate amount of energy, $\Delta E = E_2 - E_1$, can be either absorbed or emitted by the system. Plank suggests that such absorbed or emitted energy can take the form of electromagnetic radiation and that the frequency of the radiation has the simple form:

$$\nu = \Delta E / h \text{ Hz}$$

i.e.

$$\Delta E = h\nu \text{ joules}$$

Where we express our energies E in terms of the joule, and h is a universal constant-Planck's constant. This suggestion has been more than amply confirmed by experiment.

The significance of this is that if we take a molecule in state 1 and direct on to a beam of radiation of a single frequency ν (monochromatic radiation), where $\nu = \Delta E/h$, energy will be absorbed from the beam and the molecule will jump to state 2. A detector placed to collect the radiation after its interaction with the molecule will show that its intensity has decreased. Also if we use a beam containing a wide range of frequencies ('white' radiation), the detector will show that energy has been absorbed *only* from that frequency $\nu = \Delta E/h$, all other frequencies being undiminished in intensity. In this way we have produced a spectrum-an *absorption* spectrum.

Alternatively the molecule may already be in state 2 and may revert to state 1 with the consequent emission of radiation. A detector would show this radiation to have frequency $\nu = \Delta E/h$ only, and the *emission* spectrum so found is plainly complementary to the absorption spectrum of the previous paragraph.

The actual energy differences between the rotational, vibrational, and electronic energy levels are very small and may be measured in joules per molecule (or atom). In these units Planck's constant has the value:

$$h = 6.63 \times 10^{-34} \text{ joules s molecule}^{-1}$$

Often we are interested in the total energy involved when a gram-molecule of a substance changes its energy state: for this we multiply by Avogadro number $N = 6.22 \times 10^{23}$.

However, the spectroscopist measures the various characteristics of the absorbed or emitted radiation during transitions between energy states and often, rather loosely, uses frequency, wavelength, and wavenumber as if they were energy units. Thus in referring to 'an energy of 10cm^{-1} ', the spectroscopist means 'a separation between two energy states such that the associated radiation has a wavenumber value of 10cm^{-1} '.

The first expression is so simple and convenient that it is essential to become familiar with wavenumber and frequency energy units if one is to understand the spectroscopist's language. We shall use the symbol ϵ to use energy in cm^{-1} .

1.2 REGIONS OF THE SPECTRUM

1. Radiofrequency region : 3×10^6 - 3×10^{10} Hz; 10 m-1cm wavelength. Nuclear Magnetic Resonance (N.M.R) and Electron Spin Resonance (E.S.R) spectroscopy. The energy change involved is that arising from the reversal of spin of a nucleus or electron, and is of the order 0.001 - 10 joules/mole.
2. Microwave region: 3×10^{10} - 3×10^{12} Hz; 1cm-100 μ m wavelength. Rotational spectroscopy. Separations between the rotational levels of molecule are of the order of hundreds of joules per mole.
3. Infra-red region: 3×10^{12} - 3×10^{14} Hz; 100 μ m-1 μ m wavelength. Vibrational spectroscopy. One of the most valuable spectroscopic regions for the chemist. Separations between levels are some 10^4 joules/mole.
4. Visible and ultra-violet regions: 3×10^{14} - 3×10^{16} Hz; 1 μ m-10 nm wavelength. Electronic spectroscopy. The separations between the energies of valence electrons are some hundreds of kilojoules per mole.
5. X-ray: 3×10^{16} - 3×10^{18} Hz; 10nm-100 pm wavelength. Energy changes involving the inner electrons of an atom or a molecule, which may be of order ten thousand kilojoules.
6. γ -ray: 3×10^{18} - 3×10^{30} Hz; 100pm-1pm wavelength. Energy changes involve the rearrangement of nuclear particles, having energies of 10^9 - 10^{11} joules per gram atom.

Spectroscopy is the study of interaction of radiation with matter and imparts information regarding molecular symmetry, bond distance, bond angle, chemical properties, bond strength, intra and inter molecular spectra. The method of analyzing the spectra could be better illustrated by starting with small and simple molecules. The spectroscopic methods are very useful for studying of molecules. The spectroscopy is mainly applicable in quantum mechanics and group theory and the theory is used to solve more complex problems.

The molecules have different quantized energy levels for different motion. The interaction between the incident radiation and molecule, there should be a mechanism so that there must be some change in electrical or magnetic effects.

In quantum mechanical calculations gives the energy values of the various molecules energy levels because the experimental way is not applicable in verifying the absolute energy values. So, molecular spectroscopy is used like a tool, which is verifying the quantum mechanical calculation and it also helps to give the various possible reactions that take place.

In this region, the change in dipole moment makes the molecular vibrations. Consider the molecule carbon dioxide (CO_2). Carbon have net +ve charge and oxygen have net -ve charge. The molecules are alternatively stretched and compressed. Then the vibration is known as symmetric stretching.

The dipole moment remains zero throughout the motion and this particular vibration is thus infrared inactive. The another stretching vibration is called anti-symmetrical. In anti-symmetrical the alternation of moment is periodic and the vibration is thus infrared active. The infrared radiation is subdivided on the basis of the experimental procedure for obtaining the spectra.

1.3 THE VIBRATIONS OF POLYATOMIC MOLECULES

The IR spectrum is aroused due to absorption of light by vibrating molecules. When the nuclei of these molecules vibrate relative to each other the inter nuclei distance is changed. Thus, these molecules possess and oscillate dipole moment, which according to classical electrodynamics, emits radiations of frequency that fall in near infra-red region. Conversely, if an electromagnetic radiation falls on such molecule, then oscillating dipole interact with the incident radiation and can absorb radiation of infra-red region.

Consider a molecule containing N atoms: we can refer to the position of each atom by specifying three coordinates (eg. The x, y , and z Cartesian coordinates). Thus the total number of coordinate values is $3N$ and we say the molecule has $3N$ degrees of freedom since each coordinate value may be specified quite independently of the others. However, once all $3N$ coordinates have been fixed, the bond distances and bond angles of the molecule are also fixed and no further arbitrary specifications can be made.

Now the molecule is free to move in three-dimensional space, as a whole, without change of shape. We can refer to such movement by noting the position of its center of gravity at any instant-to do this requires a statement of three coordinate values. This translational movement uses three of the $3N$ degrees of freedom leaving $3N-3$. In general, also, the rotation of a non-linear molecule can be resolved into components about three perpendicular axes. Specification of these axes also requires three degrees of freedom, and the molecule is left with $3N-6$ degrees of freedom. The only other motion allowed to it is internal vibration, so we know immediately that a non-linear N -atomic molecule can have $3N-6$ different internal vibrations:

If, on the hand, the molecule is linear, there is no rotation about the bond axis; hence only two degrees of rotational freedom are required, leaving $3N-5$ degrees of vibrational freedom-one more than in the case of a non-linear molecule:

In both cases, since an N -atomic molecule has $N-1$ bonds(for acyclic molecules) between its atoms, $N-1$ of the vibrations are bond-stretching motions; the other $2N-5$ (non-linear) or $2N-4$ (linear) are bending motions. Let us look briefly at examples of these rules. First, we see that for a diatomic molecule (per force linear) such as we have already considered in $N=2$, $3N-5=1$ and thus there can be only one fundamental vibrations. Note, however that the $3N-5$ rule says nothing about the presence, absence or intensity of vibrations.

The instrument that determines the absorption spectrum for a compound is called an infra-red spectrometer or IR spectrophotometer. Two types of infra-red spectrometer are in common use: dispersive and Fourier transform instruments. Both of these types of instruments provide spectra of compounds in the common range of 4000 to 400 cm^{-1} .

1.4 ENERGY LEVELS

Vibrational energies, like other molecular energies are quantized and the permitted vibrational energies for any particular system can be calculated by Schrödinger equation. The Eigen values for the energy of a linear harmonic oscillator are of the type:

$$Ev = (v + 1/2)hc\omega$$

Where v is the vibrational quantum number equal to zero or an integer, and ' ω ' is the vibrational frequency of the oscillator expressed in wave numbers.

1.5 HARTREE-FOCK THEORY

Hartree's method was to write a conceivable approximate polyelectronic wave function for an atom as the product of one-electron wave functions:

$$\psi_0 = \varphi_0(1)\varphi_0(2) \dots \varphi_0(n)$$

This function is called Hartree product. Here ψ_0 is a function of the coordinats of all the electrons in an atom, $\varphi_0(1)$ is a function of the coordinats of electron 1, $\varphi_0(2)$ is a function of the coordinates of electron 2, etc. The initial guess wave function is φ_0 which our zeroth approximation to the total wavefunction is ψ . To apply the Hartree process we first solve for electron 1, a one electron Schrodinger equation in which the electron-electron repulsion comes from electron 1 and the

average, smeared-out electrostatic field calculated from $\varphi_0(2), \dots, \varphi_0(n)$, which is due to all other electrons.

The only moving particle in this equation is electron 1. Solving this equation gives $\varphi_1(1)$, an improved version of $\varphi_0(1)$. We next solve for electron 2, a one-electron Schrodinger equation with electron 2 move in the average field due to the electrons of $\varphi_1(1), \varphi_0(3) \dots \varphi_0(n)$, continuing to electron n moving in a field due to $\varphi_0(1), \varphi_1(2), \dots, \varphi_1(n-1)$. This completes the first cycle of calculations and gives

$$\psi_1 = \varphi_1(1)\varphi_1(2) \dots \varphi_1(n)$$

Repetition of the cycle for k times gives $\psi_2, \psi_3 \dots \psi_k$ and the energy calculated from ψ_k are essentially the same as the wavefunction and/or energy from the previous cycle. This happens when the functions for electrons 1, 2..., n are changing so little from one cycle to the next that the smeared-out electrostatic field used for the electron-electron potential would not change. At this stage the field of cycle k is essentially the same as that of k-1, which means it is “consistent with” this previous field, so that the Hartree procedure is called ‘self-consistent field procedure’, abbreviated as SCF procedure [Lewars, 2011].

Still the Hartree procedure faces two problems. One is related with the spin of the electrons, which obeys ‘Pauli exclusion principle’. In Hartree approach we acknowledge this only in an ad hoc way, simply by not placing more than two electrons in any of the component orbitals. Another problem come from the fact that electrons are indistinguishable, i.e. switching between the coordinates of two particles leave the wave function antisymmetric, but Hartree product are actually symmetric. These defects were corrected by Fock and Slater.

1.6 THE HARTREE–FOCK APPROACH

The Slater wave function is composed of spatial orbitals and spin orbitals. A spin orbital $\varphi(\text{spin})$ is the product of a spatial orbital and a spin function α or β . The spin orbitals are expressed as

$$\varphi(\text{spin}\alpha) = \varphi(\text{spatial})\alpha = \varphi(x, y, z)\alpha$$

$$\varphi(\text{spin}\beta) = \varphi(\text{spatial})\beta = \varphi(x, y, z)\beta$$

The spin functions α and β each has only one eigenvalue, $\frac{1}{2}\hbar$ or $-\frac{1}{2}\hbar$, respectively. Also the Slater wave function differs from the Hartree function not only in being composed of spin orbitals, but also in the fact that it is rather a determinant, not a simple product of one-electron functions. For a closed shell species, to construct a Slater wave function, we use each of the occupied spatial orbitals to make two spin orbitals, by multiplying the spatial orbital by α and, separately by β .

1.7 BASIS SET AND GTOs

A basis set is a set of mathematical functions, linear combination of which yield molecular orbitals. It refers to the set of non-orthogonal one-particle wave functions used to build molecular orbital, which are expanded as a linear combination of atomic orbital with the weights or coefficients to be determined.

The functions are usually, but not invariably, centered on atomic nuclei. They don't necessarily be the conventional atomic orbitals, but can be any set of mathematical functions that are convenient to manipulate and which in linear combination give useful representation of MO's. Slater and Gaussian functions are the mathematically simplest basis functions in molecular calculations.

In ab initio calculations on molecules the basis functions are almost always Gaussian functions - Gaussian Type Orbitals (GTO). Slater functions are proved to be a good approximation to atomic wave functions and would be a natural choice for ab initio basis functions, but the evaluation of certain integrals requires excessive computer time. Gaussian functions differ from Slater functions in that the exponent involves the square of distance of the electron from the point on which the function is centered, usually an atomic nucleus. An s-type Gaussian centered on nucleus A is given as

$$g_A = a_A e^{-\alpha_A |r-r_A|^2}$$

where,

$$|r - r_A|^2 = (x - x_A)^2 + (y - y_A)^2 + (z - z_A)^2$$

1.8 GAUSSIAN 09

Gaussian 09 is the latest version of the Gaussian series of electronic structure programs, used by chemists, chemical engineers, biochemists, physicists and other scientists worldwide. Gaussian 09 predicts the energies, molecular structures, vibrational frequencies and molecular properties of molecules and reactions in a wide variety of chemical environments. Gaussian 09 can predict a variety of spectra including IR and Raman, NMR, UV/Visible. Gaussian 09's models can be applied to both stable species and compounds which are difficult or impossible to observe experimentally (e.g., short-lived intermediates and transition structures).

Gaussian 09 computes the most important tensors which contribute to hyperfine spectra. Calculations can suggest regions in which to look for transitions, which can make experiments more efficient. Theoretical results are also useful for making spectral assignments for observed peaks, which can be difficult or impossible to determine solely from the raw experimental data.

Computed tensors can also be combined with observations in fitting operations. Using computations to aid in interpreting and fitting observed results should make non-linear molecules as amenable to study as linear ones. Accurate predictions of ΔG are vital to understanding many chemical reactions. Gaussian 09 offers a variety of very accurate energy methods for predicting thermochemical quantities.

CHAPTER 2

2.1 LITERATURE REVIEW

Aniline is a molecule of research interest. Vellaisamy M. et al reported the Oxidation Kinetics Of Aniline By Zinc Dichromate. The kinetics of oxidation of aniline has been investigated in aqueous acetic acid medium spectrophotometrically. The reaction shows unit order dependence each with respect to oxidant and substrate. The lowering dielectric constant of reaction medium increases the reaction rate significantly. The reaction does not form the polymerization which indicates the absence of free radical intermediate in the oxidation reaction. The oxidation of aniline yielded the corresponding azobenzene.

Role of Triplet States in the Photodynamics of Aniline was reported by Aleksandr O. et al. They studied the mechanism behind the remarkable photostability of biologically relevant chromophores, specifically focusing on aniline. The singlet manifold's efficient energy dissipation and subsequent non-radiative relaxation prevent the harmful effects of ultraviolet radiation. Recent experiments have shown triplet participation, but the mechanism remains unclear. Using theory and electronic structure capabilities.

The paper demonstrates that the population loss of aniline in the triplet manifold is dominated by spin-conserving dissociation in the T₁ state. The paper also highlights the potential of MC-PDFT as a valuable tool for probing the kinetics of excited electronic states, including spin-allowed and spin-forbidden relaxation mechanisms.

A few more recent publications discusses the Synthesis and Screening of Olefin Polymerization Catalysis using Two Aniline Derivatives and Folding and Duplex Formation in Sequence-Defined Aniline Benzaldehyde Oligoarylacetylenes.

CHAPTER 3

3.1 COMPUTATIONAL METHOD

Calculations of the title compound were carried out with Gaussian09 program using the B3LYP/6-31G basis sets to predict the molecular structure and vibrational wave numbers.. Harmonic vibrational wave numbers are calculated using the analytic second derivatives to confirm the convergence to minima on the potential surface. Then frequency calculations were employed to confirm the structure as minimum points in energy. The optimized structure of benzene(Fig:4.1) and aniline(Fig:4.2) and parameters corresponding to optimized geometry of the title compound are given in Tables(4.3 to 4.6). The absence of imaginary wavenumbers on the calculated vibrational spectrum confirms that the structure deduced corresponds to minimum energy. The assignments of the calculated wave numbers are aided by the animation option of GAUSSVIEW program, which gives a visual presentation of the vibrational modes. The potential energy distribution (PED) is calculated with the help of GAR2PED software package.

GaussView is a graphical user interface (GUI) for Gaussian. It allows users to build and visualize molecular structures, and to set up and run Gaussian calculations. GaussView also provides tools for analyzing molecular properties and visualizing results.

Gaussian is a software package that is widely used for computational chemistry. It uses quantum mechanical methods to calculate molecular properties and energies. Gaussian is used to study a variety of chemical systems, including small molecules, proteins, and materials.

Using Gauss view the molecule structure can be drawn by the following steps

Open gauss view -> view -> builder -> ring fragment -> select the structure -> click on the gaussview interface.

To add symbols and labels

View -> enable labels and symbols

Here we draw benzene structure and aniline structure , the aniline structure is benzene with one nitrogen atom . ie Open gauss view -> view -> builder -> ring fragment -> select the benzene structure -> click on the gaussview interface -> builder -> element fragment -> select the atom(nitrogen) -> click on any atom of benzene.

To obtain the frequency and vibration

Right click on molecule -> calculate -> Gaussian calculation setup

- Job type -> Opt and Freq -> enable compute raman
- Method -> basis set -> 6-31G

Submit->save in a folder as .chk extension file

Open the saved file on gauss view

Results -> vibrations

With in the folder we get a text file which contains the whole geometrical parameters such as bond angle, bond length...etc.

To obtain the ESP(electrostatic potential) distribution

Results -> surfaces/contour

- Cubes -> new cube -> type -> total density -> OK
- Surfaces -> new mapped surface -> type -> ESP -> OK

To obtain the HOMO and LUMO Values

Edit -> MOs

- Visualize -> add type -> HOMO/LUMO -> update

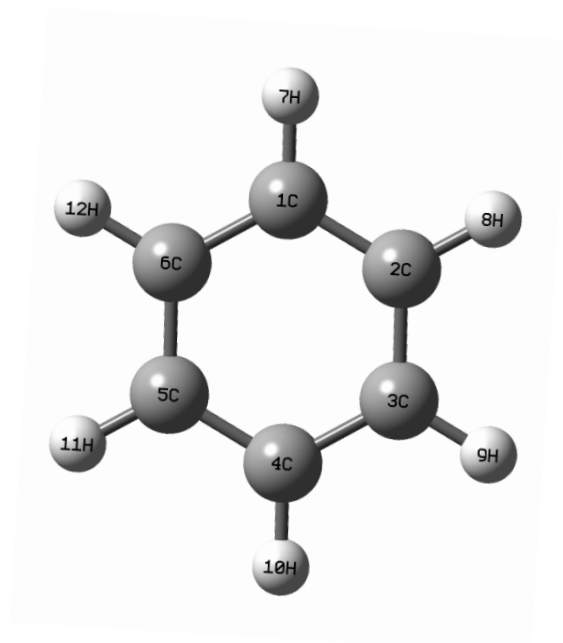
The obtained HOMO/LUMO values are multiplied with 27.1969 to convert to eV

CHAPTER 4

4.1 RESULT AND DISCUSSIONS

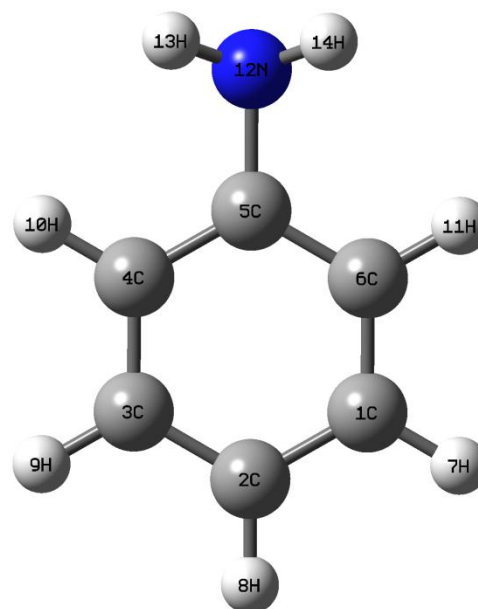
4.2 MOLECULAR STRUCTURE

Optimized Structure of Molecules



Benzene molecule structure

Figure 4.1



Aniline molecule structure

Figure 4.2

Benzene and aniline are both aromatic compounds that contain a six-membered carbon ring with alternating double bonds (known as an aromatic ring).

The optimized structure of benzene is a planar hexagon with all carbon-carbon bond lengths being equal and intermediate between a single bond and a double bond. Each carbon atom is also bonded to one hydrogen atom. The electrons in the double bonds are delocalized over the entire ring, giving it stability and aromaticity.

Aniline, on the other hand, is an aromatic amine that contains a benzene ring and an amino group (-NH₂) attached to one of its carbons. The optimized structure of aniline is similar to that of benzene, but with an additional nitrogen atom and a hydrogen atom attached to the nitrogen. The amino group attached to the benzene ring is a substituent that can participate in various chemical reactions.

Substitution of NH₂ in Benzene molecule

With the substitution of NH₂ the benzene ring, the C₁-C₂ bond length of the benzene was increased to 1.3970Å. Similarly the C₁ - C₆ bond length was increased to 1.3971Å. Likewise the C₁- H₇ bond length of the benzene was increased to 1.3802Å. Also there are variations in bond angles of the benzene with the substitution of NH₂, the bond angle of C₂- C₁ - C₆ were increased by 1°. Comparably the bond angle of C₂- C₁- H₇ were increased by 1°, the bond angle of C₁ - C₂- H₈ were decreased by 1°, the bond angle of C₃- C₂- H₈ were decreased by 1°, the bond angle of C₄- C₃- H₉ were decreased by 1°, the bond angle of C₃- C₄- C₅ were decreased by 1°, the bond angle of C₃-C₄- H₁₀ were increased by 1°, the bond angle of C₅- C₄ - H₁₀ were increased by 1°, the bond angle of C₄- C₅ - C₆ were increased by 1°, the bond angle of C₆- C₅ - H₁₁ were decreased by 1°, the bond angle of C₁ - C₆- H₁₂ were increased by 1°, the bond angle of C₅- C₆- H₁₂ were decreased by 1°.

Vibrational assignment

The maximum number of potentially active observable fundamentals of the non-linear molecule, which contains N atoms, is equal to (3N-6) apart from three translational and three rotational degrees of freedom. Here, the molecule has 14 atoms and therefore contains 36 normal modes of vibration. Vibrational spectral assignments have been performed on the recorded FT-IR spectrum based on the theoretical predicted wavenumbers using HF method. The spectral assignments of selective modes were tabulated.

C-H Vibrations:

The C-H stretching vibration in substitution with NH₂ will be shifted higher frequency due to an increase in the polarity of the C-H bond. This shift is due to an increase in the electrostatic interaction between the hydrogen and nitrogen atoms, resulting in a stronger bond. The frequency of the C-H stretching vibration will be higher for a C-H bond with an NH₂ substituent than for a C-H bond without any substituent.

The C-H stretching usually lies in the region of 2800–3400 cm⁻¹. The calculated C-H stretching vibrational modes of the molecule were obtained at the range from 3388 to 3341 cm⁻¹. The observed peaks at 3366, 3338 cm⁻¹ in FT-IR spectrum. The bands due to the C-H bending vibrations are observed in the region of 1300–1600 cm⁻¹. The calculated C-H bending vibrations coupled with other vibrational modes in the in-plane were obtained at 1325 ,1288 and 1140 cm⁻¹. Correspondingly, the peaks were observed at 1288 cm⁻¹ in FT-IR spectrum.

Also in the out of plane the calculated C-H bending vibrations coupled with other vibrational modes were obtained at 1163, 1132, 1032, 956, 877, 792 and 583 cm^{-1} . Correspondingly, the peaks were observed at 1162, 1035 and 792 cm^{-1} in FT-IR spectrum.

C-N Vibrations:

The C-N stretching usually lies in the region of 1400–1200 cm^{-1} . In the present work, the calculated C-N stretching vibrational modes coupled with certain other vibrations were obtained at 1430 cm^{-1} . The corresponding modes were observed at 1431 cm^{-1} in FT-IR spectrum. The C-N bending vibrational mode was observed at 488 cm^{-1} theoretically. The calculated C-N bending vibrations coupled with other vibrational modes in the in-plane were obtained at 416 cm^{-1} . Also in the out of plane the calculated C-N bending vibrations coupled with other vibrational modes were obtained at 249 cm^{-1} . Correspondingly, the peaks were observed at 251 cm^{-1} in FT-IR spectrum.

N-H2 Vibrations:

The N-H2 stretching usually lies in the region of 3100–3500 cm^{-1} . In the present work, the calculated N-H2 stretching vibrational modes coupled with certain other vibrations were obtained at 4004, 3869 cm^{-1} . The corresponding modes were observed at 4005, 3869 cm^{-1} in FT-IR spectrum. The C-N bending vibrational mode was observed at 1300 – 1600 cm^{-1} theoretically. The calculated C-N bending vibrations coupled with other vibrational modes in the in-plane were obtained at 1842, 1134 cm^{-1} . Correspondingly, the peaks were observed at 1844 cm^{-1} in FT-IR spectrum. Also in the out of plane the calculated C-N bending vibrations coupled with other vibrational modes were obtained at 383, 346 cm^{-1} . Correspondingly, the peaks were observed at 387 cm^{-1} in FT-IR spectrum.

4.3 IR SPECTRUM OF MOLECULES

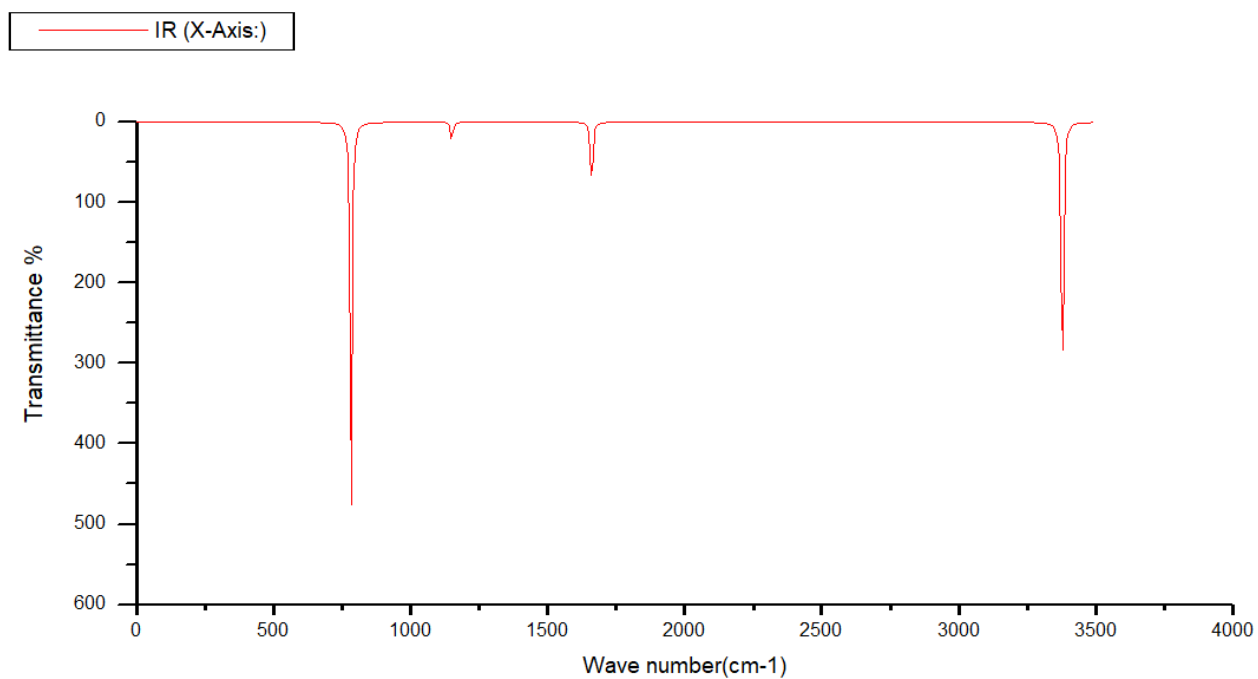


Figure 4.3 : IR Spectrum Of Benzene Molecule

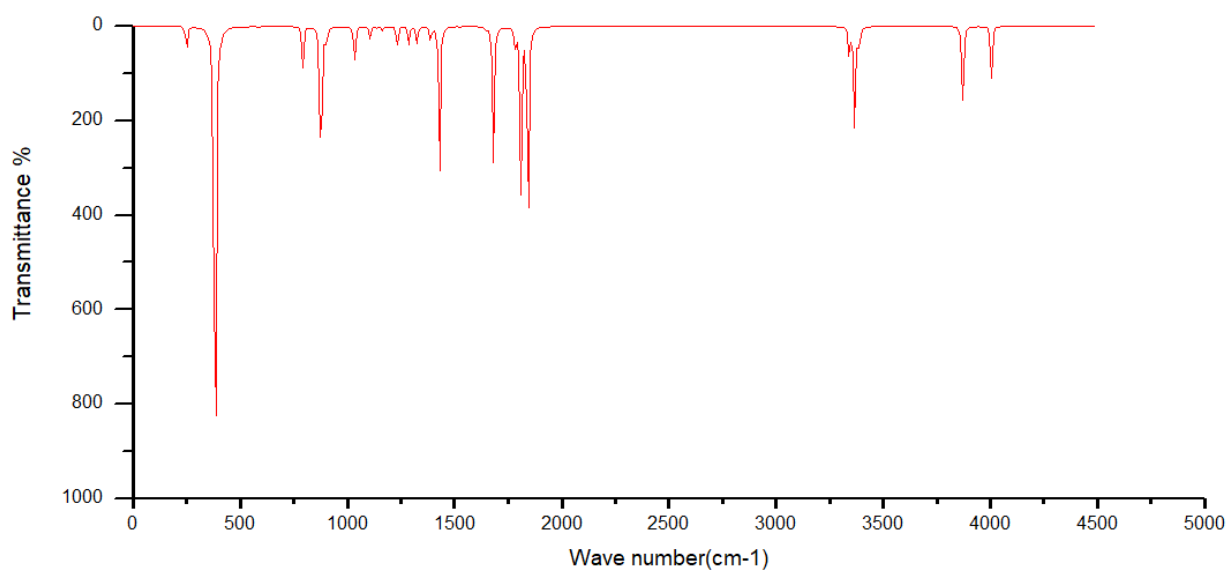


Figure 4.4 : IR Spectrum Of Aniline Molecule

VIBRATIONAL ASSIGNMENTS OF MOLECULES

Vibrational Assignments of Benzene Molecule

SL. No	ASSIGNMENT	WAVE NUMBER	IR SPECTRA (cm ⁻¹)
30	ν CH	3393	
29	ν CH	3378	3381
28	ν CH	3378	3381
27	ν CH	3358	
26	ν CH	3358	
25	ν CH	3347	
24	ν ring	1795	
23	ν ring	1795	
22	δ CH	1662	1659
21	δ CH	1662	1659
20	δ CH	1538	
19	ν ring , δ CH	1384	
18	δ CH	1320	
17	δ CH	1319	
16	ν ring , δ CH	1270	
15	γ CH	1194	
14	δ CH	1150	1147
13	δ CH	1150	1147
12	γ CH	1147	
11	γ CH	1147	
10	δ ring	1138	
9	ν ring (ring breathing)	1088	
8	γ CH	993	
7	γ CH	993	
6	γ ring	807	
5	γ CH	783	784
4	δ ring	689	
3	δ ring	689	
2	γ ring	465	
1	γ ring	465	

Table 4.1 : Vibrational Assignments of Benzene

Vibrational Assignments of Aniline Molecule

SL.No	ASSIGNMENT	WAVE NUMBER	IR SPECTRA (cm ⁻¹)
36	νNH_2	4005	4005
35	νNH_2	3871	3869
34	νCH	3388	
33	νCH	3367	3366
32	νCH	3360	3366
31	νCH	3343	3338
30	νCH	3341	3338
29	δNH_2	1842	1844
28	ν ring	1809	1810
27	ν ring	1785	1782
26	ν ring , νCN , δNH_2	1681	1683
25	ν ring , δNH_2	1649	
24	δCH , δNH_2	1523	
23	νCN	1430	1431
22	ν ring , δNH_2	1388	1387
21	δCH	1325	
20	δCH	1288	1288
19	δCH , δNH_2	1231	1233
18	γCH	1163	1162
17	δCH	1140	
16	δNH_2	1134	
15	γCH	1132	
14	ν ring (breathing)	1106	1106
13	γCH	1032	1035
12	γCH	956	
11	νCH , νCN	903	
10	γCH	877	
9	γCH	792	792
8	ν ring	705	
7	ν ring	593	
6	γCH	583	
5	γ ring	470	
4	δCN	416	
3	γNH_2	383	387
2	γNH_2	346	
1	γCN	249	252

Table 4.2 : Vibrational Assignments of Aniline

4.4 GEOMETRICAL PARAMETERS

BENZENE

BOND	BOND LENGTH(Å)
C ₁ - C ₂	1.3886
C ₁ - C ₆	1.3885
C ₁ - H ₇	1.0732
C ₂ - C ₃	1.3885
C ₂ - H ₈	1.0732
C ₃ - C ₄	1.3886
C ₃ - H ₉	1.0732
C ₄ - C ₅	1.3885
C ₄ - H ₁₀	1.0732
C ₅ - C ₆	1.3886
C ₅ - H ₁₁	1.0732
C ₆ - H ₁₂	1.0732

Table: 4.3

ANILINE

BOND	BOND LENGTH(Å)
C ₄ - C ₅	1.3970
C ₅ - C ₆	1.3971
C ₅ - N ₁₂	1.3802
C ₄ - C ₃	1.3845
C ₄ - H ₁₀	1.0744
C ₃ - C ₂	1.3880
C ₃ - H ₉	1.0738
C ₂ - C ₁	1.3880
C ₂ - H ₈	1.0724
C ₁ - C ₆	1.3845
C ₁ - H ₇	1.0738
C ₆ - H ₁₁	1.0744
N ₁₂ - H ₁₃	0.9890
N ₁₂ - H ₁₄	0.9890

Table: 4.4

BOND	BOND ANGLE(°)
C ₂ - C ₁ - C ₆	120
C ₂ - C ₁ - H ₇	120
C ₆ - C ₁ - H ₇	120
C ₁ - C ₂ - C ₃	120
C ₁ - C ₂ - H ₈	120
C ₃ - C ₂ - H ₈	120
C ₂ - C ₃ - C ₄	120
C ₂ - C ₃ - H ₉	120
C ₄ - C ₃ - H ₉	120
C ₃ - C ₄ - C ₅	120
C ₃ - C ₄ - H ₁₀	120
C ₅ - C ₄ - H ₁₀	120
C ₄ - C ₅ - C ₆	120
C ₄ - C ₅ - H ₁₁	120
C ₆ - C ₅ - H ₁₁	120
C ₁ - C ₆ - C ₅	120
C ₁ - C ₆ - H ₁₂	120
C ₅ - C ₆ - H ₁₂	120

Table: 4.5

BOND	BOND ANGLE(°)
C ₄ - C ₅ - C ₆	121
C ₄ - C ₅ - N ₁₂	121
C ₆ - C ₅ - N ₁₂	120
C ₅ - C ₄ - C ₃	120
C ₅ - C ₄ - H ₁₀	119
C ₃ - C ₄ - H ₁₀	119
C ₄ - C ₃ - C ₂	120
C ₄ - C ₃ - H ₉	120
C ₂ - C ₃ - H ₉	119
C ₃ - C ₂ - C ₁	119
C ₃ - C ₂ - H ₈	121
C ₁ - C ₂ - H ₈	121
C ₂ - C ₁ - C ₆	121
C ₂ - C ₁ - H ₇	120
C ₆ - C ₁ - H ₇	119
C ₅ - C ₆ - C ₁	120
C ₅ - C ₆ - H ₁₁	121
C ₁ - C ₆ - H ₁₁	119
C ₅ - N ₁₂ - H ₁₃	121
C ₅ - N ₁₂ - H ₁₄	118

Table: 4.6

4.5 ELECTRIC DIPOLE MOMENT

Electric dipole moment is the measure of the electrical polarity of a system of charges. In physics the electric dipole moment is a measure of the separation of the positive and negative electrical charges within a system, that is, a measure of the systems overall polarity. The electric field strength of the dipole is proportional to the magnitude of dipole moment. The SI units for electric dipole moment are coulomb-meter, however the most commonly used units is the Debye(D).

Theoretically an electric dipole is defined by the first-order term of the multipole expansion and consists of two equal and opposite charges infinitely close together. This is unrealistic as real dipoles have separated charge. However, because the charge separation is very small compared to everyday lengths, the error introduced by treating real dipoles like they are theoretically perfect is usually negligible. The direction of dipole is usually defined from the negative charge towards the positive charge.

Often in physics the dimension of a massive object can be ignored and can be treated as a point like object ie; a point particle. Point particles with electric charge are referred to as point charges. Two point charges, one with charge $-q$ separated by a distance d , constitute an electric dipole (a special case of an electric multipole). For this case, the electric dipole moment has a magnitude and is directed from the negative charge to the positive one. Some authors may split d in half and use $s=d/2$ since this quantity is the distance between either charge or the center of the dipole, leading to a factor of two in the definition.

A stronger mathematical definition is to use vector algebra, since a quantity with magnitude and direction, like the dipole moment of two point charges, can be expressed in vector form.

Where d is the displacement vector pointing from the negative charge to the positive charge. The electric dipole moment vector p also points from the negative charge to the positive charge.

An idealization of this two charge system is the electrical point dipole consisting of two (infinite) charges only infinitesimally separated, but with a finite p .

For Aniline structure H.F. calculation gives:

$$\mu_x = -0.2868272$$

$$\mu_y = 0.4967161$$

$$\mu_z = -0.0004311$$

The dipole moment is given by:

$$\mu_{\text{total}} = (\mu_x^2 + \mu_y^2 + \mu_z^2)^{\frac{1}{2}} = 0.573583 \text{ Debye}$$

4.6 FRONTIER MOLECULAR ORBITALS

The HOMO-LUMO energies are popular quantum mechanical descriptors which play a major role in governing wide range of chemical interactions. The Frontier Molecular Orbital gives an insight about the reactivity of the molecule and the active site can be demonstrated by the distribution of frontier orbital. The HOMO-LUMO frontier orbital compositions of benzene and aniline molecule were calculated with the Hatree Fock /6-31G orbital energy level are shown in Fig:4.5 and Fig:4.6 respectively. The differences between the orbital energies were shown in table: 4.7. The energy values of HOMO orbital and LUMO orbital of aniline were lying at an energy value of 7.628 eV and 4.165 eV and for the benzene molecule the energy values of HOMO orbital and LUMO orbital were lying at an energy value of 9.081 eV and 4.043 eV. The HOMO-LUMO energy gap for the aniline molecule was obtained at 5.038 eV and for the benzene molecule the energy gap is obtained at 3.463 eV. The HOMO-LUMO energy gap is related to the excitation energy of the molecule.

The concept of linear combination can be applied to polyatomic molecules and the resulting MOs are delocalized over the entire molecule. These linear combination lead to bonding, anti-bonding or non-bonding MOs. The HOMO (highest occupied molecular orbital) and LUMO (lowest unoccupied molecular orbital) are known as frontier molecular orbitals. The ionization energy (I) for an orbital is defined as the energy needed to remove an electron from the orbital to infinite separation, while the electron affinity (A) of an orbital is the energy released when the orbital accepts an electron from infinity [Koopman 1934].

$$I = -E(\text{HOMO}), \quad A = -E(\text{LUMO})$$

The orbital energy of the highest occupied MO (HOMO) is approximately equal to the ionization energy and the orbital energy of the lowest unoccupied MO (LUMO) is approximately equal to the electron affinity [Larsson 2012]. E(HOMO) is obtained as 9.30 eV and E(LUMO) is obtained as 4.98 eV.

HOMO AND LUMO OF BENZENE MOLECULE

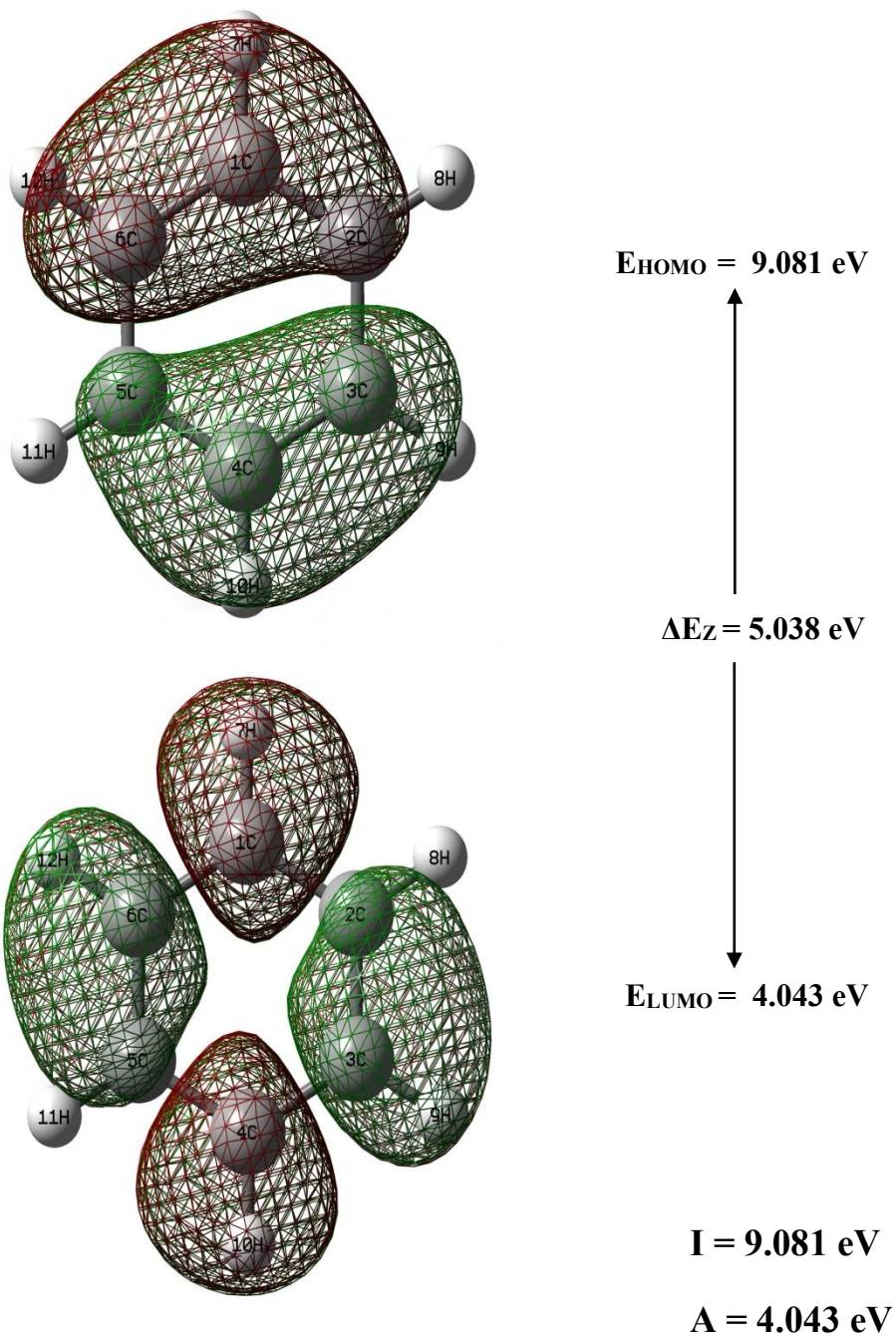


Figure 4.5: HOMO and LUMO of Benzene

HOMO AND LUMO OF ANILINE MOLECULE

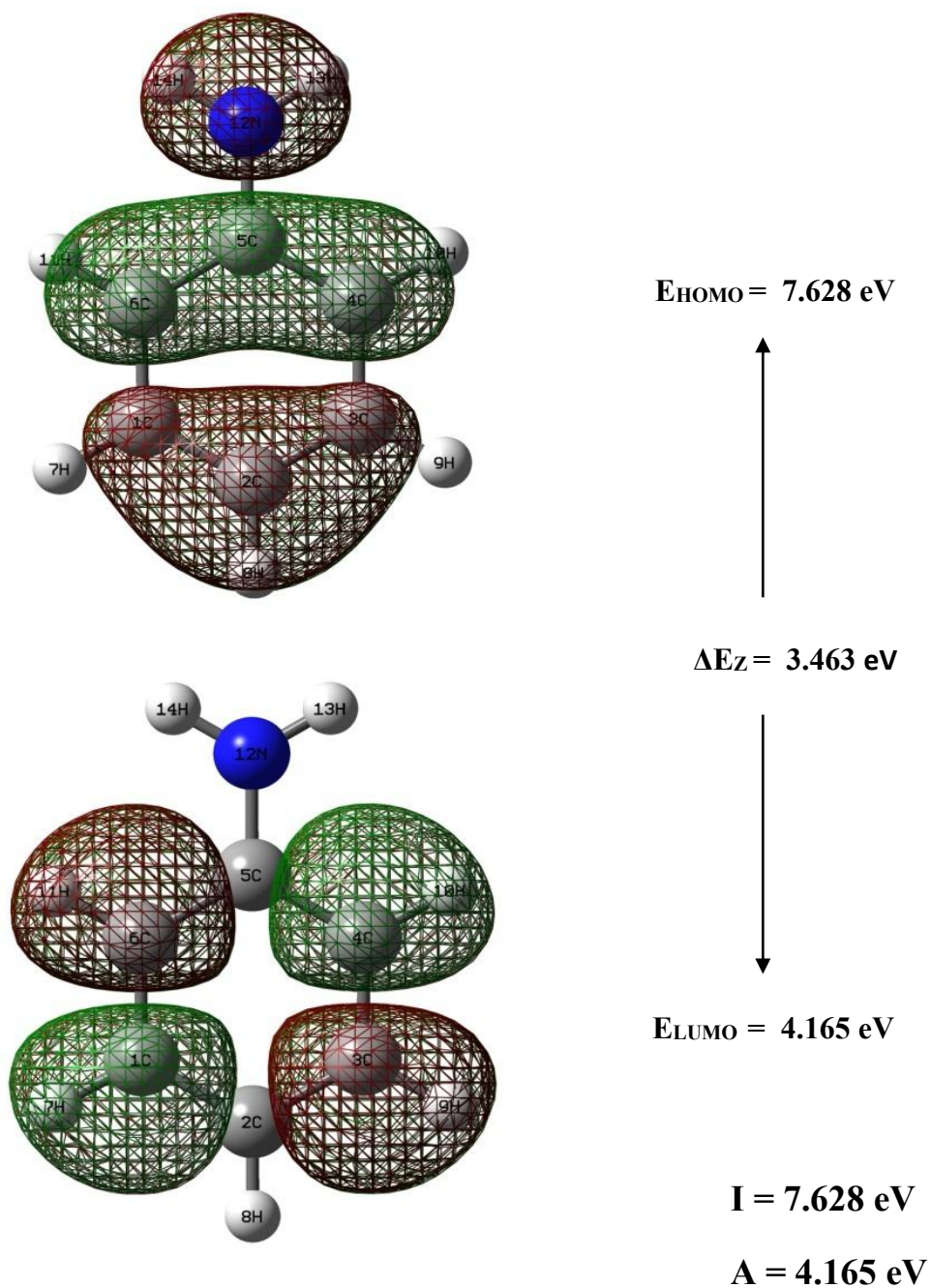


Figure 4.6: HOMO and LUMO of Aniline

Molecule	E_{HOMO} (eV)	E_{LUMO} (eV)	Ionization Energy (I)	Electron Affinity (A)	Magnitude ΔE_z (eV)
Aniline	7.628	4.165	7.628	4.165	3.463
Benzene	9.081	4.043	9.081	4.043	5.038

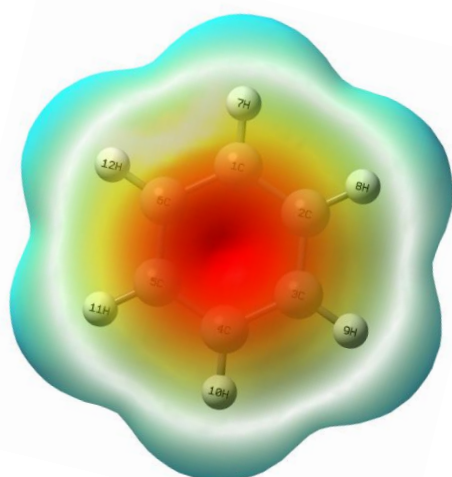
Table: 4.7

4.7 ELECTROSTATIC POTENTIAL OF MOLECULES

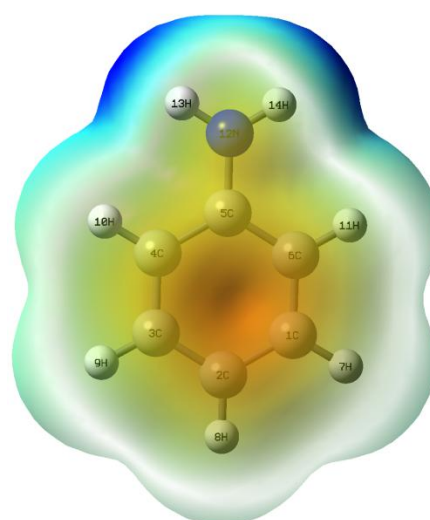
For a molecular charge distribution, which is essentially a collection of positive discrete nuclear charges Z_A and a continuous negative electron density distribution described by $\rho(\mathbf{r})$, the molecular electrostatic potential generated is given in atomic units as

$$V(\mathbf{r}) = \sum_A \frac{Z_A}{|\mathbf{r}-\mathbf{r}_A|} - \int \frac{\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} d^3\mathbf{r}'$$

The first term is the bare-nuclear potential, which is always non-negative. The second term is the negative potential, engendered by the continuous electron charge density. The resultant total MEP thus generated can attain positive as well as negative values through zero.



Benzene



Aniline

Figure 4.7 : Molecular Electrostatic Potential

The red and yellow region signifies electropositive region; the blue region signifies electronegative region. For Benzene the electropositive region is above the ring while for the aniline the electronegative region is near NH_2 . The negative (red and yellow) regions of MEP were related to electrophilic reactivity and the positive (blue) regions to nucleophilic reactivity.

CHAPTER 5

5.1 SUMMARY & CONCLUSION

The substitution effect on the benzene ring was analyzed with Gaussian computation. The IR spectra of the both the compounds were compared. The change in the bond length and bond angles with the NH₂ substitution were noted.. The molecular electrostatic potential plot shows regions of electrophilic activities for substrate binding. The Frontier Molecular Orbital studies were done to know about the HOMO and LUMO energies. The HOMO-LUMO band gap gives information about the excitation energy and frequency of the compounds. The HOMO-LUMO plot was used to identify the electron distribution and the energy gap suggests the molecules(aniline, benzene)has many uses in a lot of commercial products and industrial applications.

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